INCH-POUND

MIL-C-19978/19B
18 April 1995
SUPERSEDING
MIL-C-19978/19A
7 April 1975

MILITARY SPECIFICATION SHEET

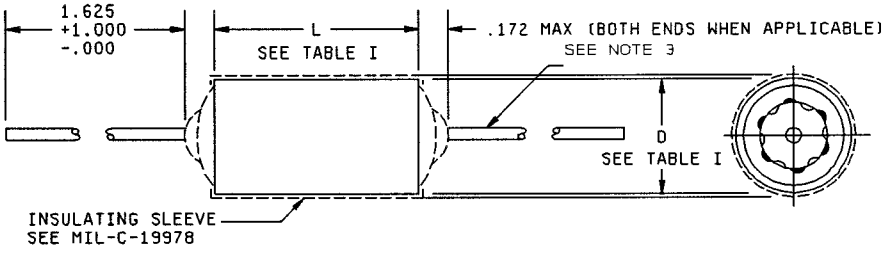
CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL TUBULAR (INSULATED) (HERMETICALLY SEALED IN METAL CASES), STYLE CQ05

INACTIVE FOR NEW DESIGN
AFTER 30 August 1974

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-19978

(B)



Inches	mm	Inches	mm
.001	0.03	.312	7.92
.004	0.10	.400	10.16
.025	0.64	.562	14.27
.032	0.81	1.000	25.40
.172	4.37	1.438	36.53
.235	5.97	1.625	41.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of tinned solid wire, .025 (No. 22 AWG) for cases .235 and .312 in diameter; and .032 (No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.
4. Capacitors with dimension L of 1.438 or D of .562 and larger, are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as a wrap-around band. The supporting device will not be supplied with the capacitor.
5. Lead length may be a minimum of 1-inch long for use in tape and reel packaging when specified in the ordering data.

FIGURE 1. Style CQ05 capacitors.

(B) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

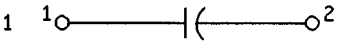
Dimensions and configuration: See figure 1.

Part or Identifying Number (PIN): See table

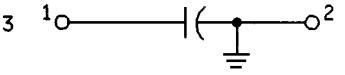
Case material: Nonmagnetic (end seal may be of magnetic material).

Dielectric material: Paper and polyethylene terephthalate (K).

CIRCUIT DIAGRAM - 1



3



Terminals: Axial-wire lead (see figure 1).
Rated voltage: See table II.

Rated temperature: -65°C to +125°C.

Capacitance (Cap.Δ) (nom): See table II.

Capacitance tolerance: -10 percent (symbol K).

Vibration grade: 3.

Dissipation factor (DF) (max): 1.0 percent.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Sleeving: In accordance with MIL-C-19978.

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Terminal to terminal: See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Immersion:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°, -0°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

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TABLE II. Style CQ05 capacitors.

PIN 1/	DC voltage rating	Capacitance	Case dimensions (in inches)		
			D +.010 -.020	L ±.031	
				Circuit 1	Circuit 3
	Volts	µF			
CQ05A-KB273K3	100	0.027	0.235	0.812	0.750
CQ05A-KB333K3	100	0.033	0.235	0.812	0.750
CQ05A-KB823K3	100	0.082	0.312	0.938	0.875
CQ05A-KB104K3	100	0.100	0.312	0.938	0.875
CQ05A-KB124K3	100	0.120	0.400	0.938	0.875
CQ05A-KB154K3	100	0.150	0.400	0.938	0.875
CQ05A-KB184K3	100	0.180	0.400	1.188	1.125
CQ05A-KB224K3	100	0.220	0.400	1.188	1.125
CQ05A-KB274K3	100	0.270	0.400	1.438	1.375
CQ05A-KB334K3	100	0.330	0.400	1.438	1.375
CQ05A-KB394K3	100	0.390	0.562	1.188	1.125
CQ05A-KB474K3	100	0.470	0.562	1.188	1.125
CQ05A-KB564K3	100	0.560	0.562	1.438	1.375
CQ05A-KB684K3	100	0.680	0.562	1.438	1.375
CQ05A-KB824K3	100	0.820	0.562	1.938	1.875
CQ05A-KB105K3	100	1.000	0.562	1.938	1.875
CQ05A-KC123K3	200	0.012	0.235	0.812	0.750
CQ05A-KC153K3	200	0.015	0.235	0.812	0.750
CQ05A-KC183K3	200	0.018	0.235	0.812	0.750
CQ05A-KC223K3	200	0.022	0.235	0.812	0.750
CQ05A-KC393K3	200	0.039	0.312	0.938	0.875
CQ05A-KC473K3	200	0.047	0.312	0.938	0.875
CQ05A-KC563K3	200	0.056	0.312	0.938	0.875
CQ05A-KC683K3	200	0.068	0.312	0.938	0.875
CQ05A-KC823K3	200	0.082	0.400	0.938	0.875
CQ05A-KC104K3	200	0.100	0.400	0.938	0.875
CQ05A-KC124K3	200	0.120	0.400	1.188	1.125
CQ05A-KC154K3	200	0.150	0.400	1.188	1.125
CQ05A-KC184K3	200	0.180	0.400	1.438	1.375
CQ05A-KC224K3	200	0.220	0.400	1.438	1.375
CQ05A-KC274K3	200	0.270	0.562	1.188	1.125
CQ05A-KC334K3	200	0.330	0.562	1.188	1.125
CQ05A-KC394K3	200	0.390	0.562	1.438	1.375
CQ05A-KC474K3	200	0.470	0.562	1.438	1.375
CQ05A-KC564K3	200	0.560	0.562	1.938	1.875
CQ05A-KC684K3	200	0.680	0.562	1.938	1.875
CQ05A-KC824K3	200	0.820	0.670	1.938	1.875
CQ05A-KC105K3	200	1.000	0.670	1.938	1.875
CQ05A-KE822K3	400	0.0082	0.235	0.812	0.750
CQ05A-KE103K3	400	0.010	0.235	0.812	0.750
CQ05A-KE273K3	400	0.027	0.312	0.938	0.875
CQ05A-KE333K3	400	0.033	0.312	0.938	0.875
CQ05A-KE393K3	400	0.039	0.400	0.938	0.875
CQ05A-KE473K3	400	0.047	0.400	0.938	0.875

1/ See footnote at end of table.

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TABLE II. Style CQ05 capacitors - Continued.

PIN 1/	DC voltage rating	Capacitance	Case dimensions (in inches)		
			D +.010 -.020	L ±.031	
				Circuit 1	Circuit 3
	Volts	µF			
CQ05A-KE563K3	400	0.056	0.400	1.188	1.125
CQ05A-KE683K3	400	0.068	0.400	1.188	1.125
CQ05A-KE823K3	400	0.082	0.400	1.438	1.375
CQ05A-KE104K3	400	0.100	0.400	1.438	1.375
CQ05A-KE124K3	400	0.120	0.562	1.188	1.125
CQ05A-KE154K3	400	0.150	0.562	1.188	1.125
CQ05A-KE184K3	400	0.180	0.562	1.438	1.375
CQ05A-KE224K3	400	0.220	0.562	1.438	1.375
CQ05A-KE274K3	400	0.270	0.562	1.938	1.875
CQ05A-KE334K3	400	0.330	0.562	1.938	1.875
CQ05A-KE394K3	400	0.390	0.670	1.938	1.875
CQ05A-KE474K3	400	0.470	0.670	1.938	1.875
CQ05A-KE564K3	400	0.560	0.750	2.188	2.125
CQ05A-KE684K3	400	0.680	0.750	2.188	2.125
CQ05A-KE824K3	400	0.820	1.000	1.938	1.875
CQ05A-KE105K3	400	1.000	1.000	1.938	1.875
CQ05A-KF102K3	600	0.001	0.235	0.812	0.750
CQ05A-KF122K3	600	0.0012	0.235	0.812	0.750
CQ05A-KF152K3	600	0.0015	0.235	0.812	0.750
CQ05A-KF182K3	600	0.0018	0.235	0.812	0.750
CQ05A-KF222K3	600	0.0022	0.235	0.812	0.750
CQ05A-KF272K3	600	0.0027	0.235	0.812	0.750
CQ05A-KF332K3	600	0.0033	0.235	0.812	0.750
CQ05A-KF392K3	600	0.0039	0.235	0.812	0.750
CQ05A-KF472K3	600	0.0047	0.235	0.812	0.750
CQ05A-KF562K3	600	0.0056	0.235	0.812	0.750
CQ05A-KF682K3	600	0.0068	0.235	0.812	0.750
CQ05A-KF822K3	600	0.0082	0.312	0.938	0.875
CQ05A-KF103K3	600	0.010	0.312	0.938	0.875
CQ05A-KF123K3	600	0.012	0.312	0.938	0.875
CQ05A-KF153K3	600	0.015	0.312	0.938	0.875
CQ05A-KF183K3	600	0.018	0.312	0.938	0.875
CQ05A-KF223K3	600	0.022	0.312	0.938	0.875
CQ05A-KF273K3	600	0.027	0.400	0.938	0.875
CQ05A-KF333K3	600	0.033	0.400	0.938	0.875
CQ05A-KF393K3	600	0.039	0.400	1.188	1.125
CQ05A-KF473K3	600	0.047	0.400	1.188	1.125
CQ05A-KF563K3	600	0.056	0.400	1.438	1.375
CQ05A-KF683K3	600	0.068	0.400	1.438	1.375
CQ05A-KF823K3	600	0.082	0.562	1.188	1.125
CQ05A-KF104K3	600	0.100	0.562	1.188	1.125
CQ05A-KF124K3	600	0.120	0.562	1.438	1.375
CQ05A-KF154K3	600	0.150	0.562	1.438	1.375
CQ05A-KF184K3	600	0.180	0.562	1.938	1.875
CQ05A-KF224K3	600	0.220	0.562	1.938	1.875
CQ05A-KF274K3	600	0.270	0.562	1.938	1.875
CQ05A-KF334K3	600	0.330	0.562	1.938	1.875
CQ05A-KF394K3	600	0.390	0.750	2.188	2.125
CQ05A-KF474K3	600	0.470	0.750	2.188	2.125
CQ05A-KF564K3	600	0.560	1.000	1.938	1.875
CQ05A-KF684K3	600	0.680	1.000	1.938	1.875
CQ05A-KG153K3	1,000	0.015	0.400	0.938	0.875
CQ05A-KG223K3	1,000	0.022	0.400	1.188	1.125
CQ05A-KG273K3	1,000	0.027	0.400	1.438	1.375
CQ05A-KG333K3	1,000	0.033	0.400	1.438	1.375

1/ See footnote at end of table.

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TABLE II. Style CQ05 capacitors - Continued.

PIN 1/	DC voltage rating	Capacitance	Case dimensions (in inches)		
			D +.010 -.020	L ±.031	
				Circuit 1	Circuit 3
	Volts	µF			
CQ05A-KG473K3	1,000	0.047	0.562	1.188	1.125
CQ05A-KG563K3	1,000	0.056	0.562	1.438	1.375
CQ05A-KG683K3	1,000	0.068	0.562	1.438	1.375
CQ05A-KG823K3	1,000	0.082	0.562	1.688	1.625
CQ05A-KG104K3	1,000	0.100	0.562	1.688	1.625
CQ05A-KG124K3	1,000	0.120	0.670	1.688	1.625
CQ05A-KG154K3	1,000	0.150	0.670	1.688	1.625
CQ05A-KG224K3	1,000	0.220	0.750	2.188	2.125
CQ05A-KG274K3	1,000	0.270	0.750	2.688	2.625
CQ05A-KG334K3	1,000	0.330	0.750	2.688	2.625
CQ05A-KG474K3	1,000	0.470	1.000	2.188	2.125

1/ Complete PIN shall include an additional symbol to indicate circuit.

TABLE III. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm	Inches	mm
0.010	0.25	0.812	20.62	1.688	42.88
0.020	0.51	0.875	22.23	1.875	47.63
0.031	0.79	0.938	23.83	1.938	49.23
0.235	5.97	1.000	25.40	2.125	53.98
0.312	7.92	1.125	28.58	2.188	55.58
0.400	10.16	1.188	30.18	2.625	66.68
0.562	14.27	1.375	34.93	2.688	68.28
0.670	17.02	1.438	36.53		
0.750	19.05	1.625	41.28		

Substitution data: See table IV.

NOTE: Style CQ05 capacitors are substitutable for styles CP04 and CP05 capacitors formerly covered in MIL-C-25/1K.

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TABLE IV. Substitution data.

FROM MIL-C-25/1K (CP04 and CP05)	TO MIL-C-19978/19A (CQ05)	FROM MIL-C-25/1K (CP04 and CP05)	TO MIL-C-19978/19A (CQ05)
CP0-A-KB273K-	CQ05A-KB273K3	CP0-A-KC223K-	CQ05A-KC223K3
CP0-A-KB333K-	CQ05A-KB333K3	CP0-A-KC393K-	CQ05A-KC393K3
CP0-A-KB823K-	CQ05A-KB823K3	CP0-A-KC473K-	CQ05A-KC473K3
CP0-A-KB104K-	CQ05A-KB104K3	CP0-A-KC563K-	CQ05A-KC563K3
CP0-A-KB124K-	CQ05A-KB124K3	CP0-A-KC683K-	CQ05A-KC683K3
CP0-A-KB154K-	CQ05A-KB154K3	CP0-A-KC823K-	CQ05A-KC823K3
CP0-A-KB184K-	CQ05A-KB184K3	CP0-A-KC104K-	CQ05A-KC104K3
CP0-A-KB224K-	CQ05A-KB224K3	CP0-A-KC124K-	CQ05A-KC124K3
CP0-A-KB274K-	CQ05A-KB274K3	CP0-A-KC154K-	CQ05A-KC154K3
CP0-A-KB334K-	CQ05A-KB334K3	CP0-A-KC184K-	CQ05A-KC184K3
CP0-A-KB394K-	CQ05A-KB394K3	CP0-A-KC224K-	CQ05A-KC224K3
CP0-A-KB474K-	CQ05A-KB474K3	CP0-A-KC274K-	CQ05A-KC274K3
CP0-A-KB564K-	CQ05A-KB564K3	CP0-A-KC334K-	CQ05A-KC334K3
CP0-A-KB684K-	CQ05A-KB684K3	CP0-A-KC394K-	CQ05A-KC394K3
CP0-A-KB824K-	CQ05A-KB824K3	CP0-A-KC474K-	CQ05A-KC474K3
CP0-A-KB105K-	CQ05A-KB105K3	CP0-A-KC564K-	CQ05A-KC564K3
CP0-A-KC123K-	CQ05A-KC123K3	CP0-A-KC684K-	CQ05A-KC684K3
CP0-A-KC153K-	CQ05A-KC153K3	CP0-A-KC824K-	CQ05A-KC824K3
CP0-A-KC183K-	CQ05A-KC183K3	CP0-A-KC105K-	CQ05A-KC105K3
CP0-A-KE822K-	CQ05A-KE822K3	CP0-A-KF153K-	CQ05A-KF153K3
CP0-A-KE103K-	CQ05A-KE103K3	CP0-A-KF183K-	CQ05A-KF183K3
CP0-A-KE273K-	CQ05A-KE273K3	CP0-A-KF223K-	CQ05A-KF223K3
CP0-A-KE333K-	CQ05A-KE333K3	CP0-A-KF273K-	CQ05A-KF273K3
CP0-A-KE393K-	CQ05A-KE393K3	CP0-A-KF333K-	CQ05A-KF333K3
CP0-A-KE473K-	CQ05A-KE473K3	CP0-A-KF393K-	CQ05A-KF393K3
CP0-A-KE563K-	CQ05A-KE563K3	CP0-A-KF473K-	CQ05A-KF473K3
CP0-A-KE683K-	CQ05A-KE683K3	CP0-A-KF563K-	CQ05A-KF563K3
CP0-A-KE823K-	CQ05A-KE823K3	CP0-A-KF683K-	CQ05A-KF683K3
CP0-A-KE104K-	CQ05A-KE104K3	CP0-A-KF823K-	CQ05A-KF823K3
CP0-A-KE124K-	CQ05A-KE124K3	CP0-A-KF104K-	CQ05A-KF104K3
CP0-A-KE154K-	CQ05A-KE154K3	CP0-A-KF124K-	CQ05A-KF124K3
CP0-A-KE184K-	CQ05A-KE184K3	CP0-A-KF154K-	CQ05A-KF154K3
CP0-A-KE224K-	CQ05A-KE224K3	CP0-A-KF184K-	CQ05A-KF184K3
CP0-A-KE274K-	CQ05A-KE274K3	CP0-A-KF224K-	CQ05A-KF224K3
CP0-A-KE334K-	CQ05A-KE334K3	CP0-A-KF274K-	CQ05A-KF274K3
CP0-A-KE394K-	CQ05A-KE394K3	CP0-A-KF334K-	CQ05A-KF334K3
CP0-A-KE474K-	CQ05A-KE474K3	CP0-A-KF394K-	CQ05A-KF394K3
CP0-A-KE564K-	CQ05A-KE564K3	CP0-A-KF474K-	CQ05A-KF474K3
CP0-A-KE684K-	CQ05A-KE684K3	CP0-A-KF564K-	CQ05A-KF564K3
CP0-A-KE824K-	CQ05A-KE824K3	CP0-A-KF684K-	CQ05A-KF684K3
CP0-A-KE105K-	CQ05A-KE105K3	CP0-A-KG153K-	CQ05A-KG153K3
CP0-A-KF102K-	CQ05A-KF102K3	CP0-A-KG223K-	CQ05A-KG223K3
CP0-A-KF122K-	CQ05A-KF122K3	CP0-A-KG273K-	CQ05A-KG273K3
CP0-A-KF152K-	CQ05A-KF152K3	CP0-A-KG333K-	CQ05A-KG333K3
CP0-A-KF182K-	CQ05A-KF182K3	CP0-A-KG473K-	CQ05A-KG473K3
CP0-A-KF222K-	CQ05A-KF222K3	CP0-A-KG563K-	CQ05A-KG563K3
CP0-A-KF272K-	CQ05A-KF272K3	CP0-A-KG683K-	CQ05A-KG683K3
CP0-A-KF332K-	CQ05A-KF332K3	CP0-A-KG823K-	CQ05A-KG823K3
CP0-A-KF392K-	CQ05A-KF392K3	CP0-A-KG104K-	CQ05A-KG104K3
CP0-A-KF472K-	CQ05A-KF472K3	CP0-A-KG124K-	CQ05A-KG124K3
CP0-A-KF562K-	CQ05A-KF562K3	CP0-A-KG154K-	CQ05A-KG154K3
CP0-A-KF682K-	CQ05A-KF682K3	CP0-A-KG224K-	CQ05A-KG224K3
CP0-A-KF822K-	CQ05A-KF822K3	CP0-A-KG274K-	CQ05A-KG274K3
CP0-A-KF103K-	CQ05A-KF103K3	CP0-A-KG334K-	CQ05A-KG334K3
CP0-A-KF123K-	CQ05A-KF123K3	CP0-A-KG474K-	CQ05A-KG474K3

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, MC, OS, SH, TD
Air Force - 17, 19, 99

Preparing activity:
DLA - ES

(Project 5910-1902)

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